

Semantic-aware Responsive Listener Head Synthesis

Wei Zhao
zhaoweiheap@hnu.edu.cn
Hunan University
Changsha, Hunan, China

Peng Xiao
napping@hnu.edu.cn
Hunan University
Changsha, Hunan, China

Rongju Zhang
SpallopZrj@hnu.edu.cn
Hunan University
Changsha, Hunan, China

Yijun Wang
wyjun@hnu.edu.cn
Hunan University
Changsha, Hunan, China

Jianxin Lin*
linjianxin@hnu.edu.cn
Hunan University
Changsha, Hunan, China

ABSTRACT

Audience providing proper reaction during a conversation can bring positive impact to speaker, which is significant to digital human and social agent areas. Given information sent by speaker, responsive listener head synthesis task aims to generate corresponding listener behaviours such as nodding, thinking and smiling. A common method is to build listener responsive pattern by analyzing acoustic and facial feature of speaker. However, it is hard to understand what speaker means, purely based on acoustic and facial feature since numerous message is buried in language. Traditional method may lead to similar results ignoring the diversity of input. Therefore, in this paper we presents a new Semantic-aware Responsive Listener Head Synthesis (SaRLHS) approach by considering semantic information lied in language patterns in addition to acoustic and facial feature. Besides, we implement a post-face enhancement process to increase the visual effects. Moreover, we won the People's Selection Awards and the second place on Grand Challenges of ACM 2022 conference.

CCS CONCEPTS

• Computing methodologies → Computer vision.

KEYWORDS

responsive listener head synthesis, semantic-aware

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*Corresponding author.

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1 INTRODUCTION

Recently, digital human has been implemented in many fields such as video games, films or TV shows and virtual companions. The rise of meta universe continues to boost the value of digital human with the global increasing trend of digitalization [26]. As the name suggests, a digital human is a human-like avatar that can conduct a series of human behaviors powered by artificial intelligent. When considering the design of digital human, communication is doubtlessly one of the most significant capacities of digital human since face-to-face communication is one of the most common behaviours in human's daily life [11]. It usually revolves two roles, i.e., speaker and listener, when a face-to-face conversation takes place. There are many research efforts on speaker-centric synthesis such as generating gesture according to a clip of audio material [7], outputting a lip-synchronization video with the input of speech [17], synthesizing talking face video driven by audio [4], etc.

Compared to speaker-centric synthesis, there has been much less research concerning the listener behaviours, ignoring the value of listener on a conversation. It is natural to pay more attention to speaker since speaker starts a conversation and transmits more information to outsider than listener does. However, listener is important to speaker since a communication activity can not proceed without the extent of listener. Besides, it is necessary to be a good listener particularly in digital human area as digital human often shows as an assistant. Providing proper feedback may help increasing the experience of user. Given real-time information sent by speaker, responsive head generation task tends to provide spontaneous feedback. [16] proposed a multi-modal listening system to achieve social tasks by analyzing previous analyses of listener behaviour like gaze pattern. [10] refers this type of system as Social Artificial Intelligence (Social AI). [10] devotes to building Social AI based on nonverbal interaction such as gesture, gaze, facial expression and 3D body. Both of them notice the unique status of interaction but blur the bound between listener and speaker without verbal interaction. Although [27] takes audio as signal of speaker, it ignores semantic message hidden in audio instead. Overcoming these drawbacks and combing the advantage of the method, we we presents a new Semantic-aware Responsive Listener Head Synthesis (SaRLHS) approach by considering semantic information lied in language patterns in addition to acoustic and facial feature. Main contribution of this work include:

- We propose to exploit the semantic information lied in speaker's speech for responsive listener generation.

- We propose a post-face enhancement process to increase the face rendering effects.

2 RELATED WORK

Communication plays an important role at virtual human areas since it is one of the most common activities during human daily life. Recently, there have been many research on talking head generation to imitate human talking given a driving material such as audio. Wav2Lip [17] devotes to producing accurate lip-synchronization given an audio. Unlike other works that generate lip movements on a certain image or video, Wav2Lip is capable of synthesizing lip-syncing videos in dynamic and unconstrained talking face. Speech-to-gesture [7] learns connection between gesture and speech in conversational situation. Chung et al. [4] leverage an encoder-decoder architecture to generate a talking head video of lip syncing with given audio and static portrait image. MEAD [22] aims to control emotion when proceed talking face generation. A conversation often revolves two roles, speaking roles and listening roles. However, the generation talking head mentioned before only considering the information about speaking role like audio, gesture and emotion without thinking of listening roles which provide feedback for speakers to proceed the conversation. It is obvious that listener is significant in simulating the communication situation [15, 19–21]. Different from speaker-centered video synthesis more like a translation task, listener-centered task need to understand the latent information under the visual and audio signal of speaker before providing reaction. Gillies et al. [6] construct a data driven model creating unrealistic character that can respond to a real human on a conversation. Kington et al. [12] study the importance of appropriate response for an agent in conversation. Aiming to achieving interaction with virtual agent, McKeown et al. [14] create a large database containing audio and visual records of conversation. Buschmeier et al. [3] provide rich corpus ALICO about real-time face-to-face communication. Besides, ALICO distinguish distracted listener and attentive one. Capturing signals sent by speakers such as face and gesture, Joo et al. [10] propose a nonverbal signals estimation framework focusing on gesture and head movements. Oertel et al. [16] build a generation system to synthesize various behavior of listeners. Similar to talking head synthesis, recent research on listening head generation task Joo et al. [10, 16] concentrate on listener ignoring the other role in a conversation. Inspired by Zhou et al. [27], we propose a spontaneous listening head generation receiving a streaming of visual and audio information.

3 METHOD

In our daily life, listeners need to comprehend the signals such as expressions and audio information sent by speakers before they provide feedback. Besides, listeners react to speakers at the very moment based on the information of that moment as well as information from history. Therefore, the goal of responsive listening head generation task is to predict the constant reactions of listeners according to the real-time signals from speakers. In this work, We design an architecture composed of an encoder, an estimator and a decoder to achieve the goal. The encoder aims to encode and fuse the signals of speakers. The estimator aims to predict responsive listening head motion feature vectors. The decoder aims to render

the listening head motion feature vectors into responsive listener video based on a reference image. The overview of our approach is illustrated in Fig. 1

3.1 Encoder

The encoder consists of a visual part and an acoustic part. Given a video sequence of speaker $V_T = \{v_1, \dots, v_T\}$ where time stamps range in $\{1, \dots, T\}$, the visual part focuses on regressing the 3DMM coefficients [1]. For time stamp t , we get the 3DMM coefficients $x_t = (\alpha_t, \beta_t, \epsilon_t, \gamma_t, p_t)$ where β_t and p_t represent features related to identity while α_t , ϵ_t and γ_t denote identity-dependent features.

For acoustic part, except for Mel-frequency cepstral coefficients (MFCC) $A_T = \{a_1, \dots, a_T\}$ extracted from the audio, we implement the pre-trained language model BERT [5] to extract semantic representations $S_T = \{s_1, \dots, s_T\}$ from text that recognized from the audio.

After acquiring visual and acoustic representations of speaker, a multi-modal feature fusion function f_{va} is applied to get the encoded feature of speaker:

$$e_t = f_{va}(\beta_t, p_t, a_t, s_t), \quad (1)$$

where β_t and p_t are separated from x_t .

3.2 Estimator

At each time step t , the estimator module takes the speaker's feature e_t as input and generates the listener motion feature vectors β_{t+1} and p_{t+1} . LSTM [8] is applied considering its sequential property. The procedure can be formulated as:

$$\beta_{t+1}, p_{t+1} = LSTM(c_t, e_t), \quad (2)$$

where $0 \leq t \leq T$, c_0 is initial and self-defined state applied for driving LSTM to generated the first state while c_t is current state which participates next prediction.

3.3 Decoder

The decoder is composed of a coarse grain renderer G_v and a fine grain adjuster. Inspired by coarse renderer [18], the decoder first generates a sequence of listener portrait images based on a reference listener portrait image I containing 3DMM coefficients related to identity, illumination, background, estimated expression and motion. We employ the 3DMM coefficients as the target motion descriptor p to instruct synthesizing the realistic listener frame of the conversational video. Firstly, we map descriptor p to latent vectors z via a mapping network f_m :

$$z = f_m(p). \quad (3)$$

We further transform the latent vector to z_a and z_b by affine transformation and conduct adaptive instance normalization for face i :

$$z_a, z_b = \text{transformation}(z), \quad (4)$$

$$\text{AdaIN}(x_i, z) = z_{a,i} \frac{x_i - \mu(x_i)}{\sigma(x_i)} + z_{b,i}.$$

To maintain texture and background of reference portrait image, we implement a warping network f_w to learn the coordinate offsets

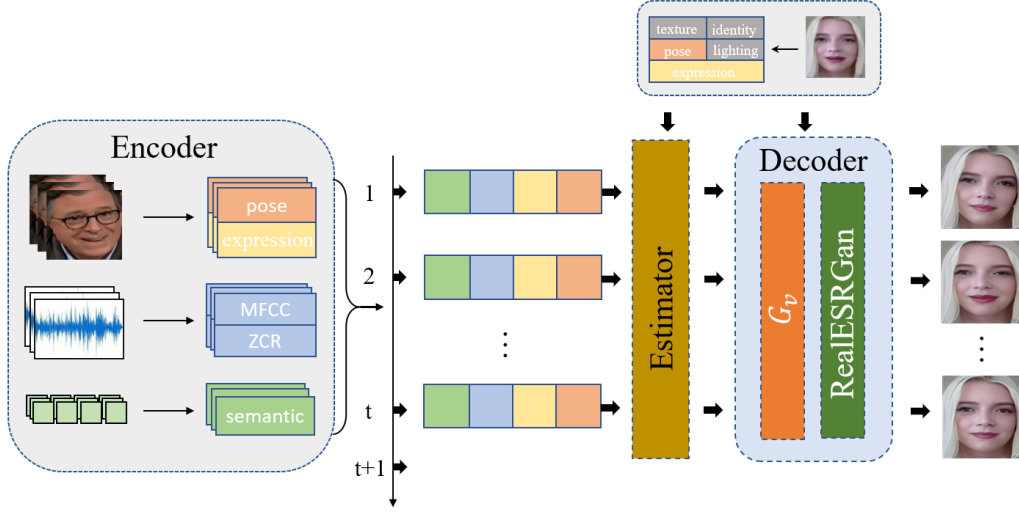


Figure 1: The framework of Semantic-aware Responsive Listener Head Synthesis (SaRLHS). 1) The left part denotes Encoder that encodes facial, acoustic and semantic feature; 2) The middle part denotes Estimator where t denotes time stamps of predicted 3dmm coefficients; 3) The right part denotes Decoder that synthesize ultimate video of listener in both coarse and fine level.

between reference portrait and target motion:

$$\begin{aligned} w &= f_w(I, z_a, z_b), \\ \hat{I}_w &= w(I), \end{aligned} \quad (5)$$

where $\hat{I}_w$ is the coarse result. The network adopts an auto-encoder framework. Affined vectors z_a and z_b inject target motions into the network with AdaIN operation.

As warping network is unable to generate unknown pixels, a generation network using similar architecture as warping network is applied to modify the warped coarse results:

$$\hat{I} = g(\hat{I}_w, I, z_a, z_b). \quad (6)$$

For fine grained adjustment, we adopt a practical restoration application, i.e., Real-ESRGAN [23], to achieve visual enhancement. The restoration network employs several residual-in-residual dense blocks (RRDB) while extending scale factor of $\times 2$ and $\times 1$. RRDB consists of multi-level residual network and dense layers without batch normalization layers, which is the same as ESRGAN [24]. For discriminator, a U-net architecture with skip connections is designed to tackle degradation problem.

3.4 Training Strategy

For estimator, we use L_2 distance to supervise the predicted expression and head motion to guarantee the inter-frame continuity [27]:

$$\begin{aligned} \mathcal{L}_{est} &= \sum_{t=2}^T \left\| \beta_t^l - \hat{\beta}_t^l \right\|_2 + \left\| p_t^l - \hat{p}_t^l \right\|_2 \\ &+ \sum_{t=2}^T w_1 \left\| \mu(\beta_t^l) - \mu(\hat{\beta}_t^l) \right\|_2 + w_2 \left\| \mu(p_t^l) - \mu(\hat{p}_t^l) \right\|_2. \end{aligned} \quad (7)$$

where $\mu(\cdot)$ measures the inter-frame changes of current frame and its adjacent previous frame.

For decoder, following Pirenderer [18], we use VGG-19 network to constrain warping and reconstruction network:

$$\begin{aligned} \mathcal{L}_{dec} &= \lambda_w \sum_i \left\| \phi_i(I_t) - \phi_i(\hat{I}_w) \right\|_1 \\ &+ \lambda_c \sum_i \left\| \phi_i(I_t) - \phi_i(\hat{I}) \right\|_1 \\ &+ \lambda_s \sum_j \left\| G_j^\phi(I_t) - G_j^\phi(\hat{I}) \right\|_1, \end{aligned} \quad (8)$$

where ϕ_i is the activation map of the i -th layer of VGG19 while G_j^ϕ is the Gram matrix constructed from activation maps ϕ_j .

The final loss function can be formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{est} + \mathcal{L}_{dec}. \quad (9)$$

4 EXPERIMENT

Training details For training, we used vico [27], a conversational video clips containing 483 video clips with the size of 384×384 . Our framework adopts AdamW [13] as optimizer with a batch size of 16, $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate is fixed value of 1×10^{-3} for 500 epochs reduced exponentially by 0.8 every 50 epochs. All the experiments are conducted on an NVIDIA GeForce 3090 with 24 GB GPUs.

Evaluation Tab 1 shows quantitative results of the evaluation. Among these evaluation, SSIM [25], CPBD [2] and PSNR [2] depict generation quality from image level while FID [9] depicts generation quality from feature level. CSIM is cosine similarity showing

Table 1: The quantitative results of baseline and ours.

	SSIM	CPBD	PSNR	FID	CSIM	ExpFD	AngleFD	TransFD
Baseline	0.509	0.114	16.416	26.205	0.213	0.149	0.070	0.063
Ours	0.495	0.514	16.780	80.538	0.205	0.161	0.054	0.056

Table 2: The quantitative results of our model in various version.

	SSIM	CPBD	PSNR	FID	CSIM	ExpFD	AngleFD	TransFD
Ours(witout BERT)	0.492	0.398	16.265	86.983	0.173	0.167	0.069	0.068
Ours(without Real-ESRGAN)	0.532	0.113	17.105	25.723	0.245	0.138	0.054	0.056
Ours	0.495	0.514	16.780	80.538	0.205	0.161	0.054	0.057

**Figure 2: a) baseline results; b) ours w/o Real-ESRGAN results c) ours w/o bert results d) our full model results**

the extent of identity preserving. Instead of evaluating the visual effects of decoder, ExpFD denotes L_1 distance between ground-truth and synthesized video which reflects feature of expression. Similar to ExpFD, AngleFD and TransFD represents L_1 distance between ground-truth and synthesized video which reflect feature of head motion. Ours are final result we submitted to track #2 of Conversational Head Generation Challenge. As we can see in the Tab 1, except for FID our results perform better than baseline or almost as good as baseline. FID works for comparing distance between two distribution which is ground-truth video and generated video. After adding Real-ESRGAN, model acquires a great promotion at visual effects as showed in Tab 2. We assume that the distribution of generated video locates further than ground-truth due to this promotion. In other words, the higher FID means the better generation quality in this case.

Ablation Study To verify the importance of added modules, we set up two additional experiments one with Bert without Real-ESRGAN and one with Real-ESRGAN without Bert. The ablation

experiment results are show in Tab 2. Without Bert, the model perform worse in all evaluation rules. Although acquiring better evaluation in all evaluation after removing Real-ESRGAN modul, our model make a huge progress at visual effects as showed in the Fig 2.

5 CONCLUSION

In this paper, we develop a LSTM-based listening head generator that aims to synthesize corresponding listener behaviours in a time-various manner given a video of speaker. Instead of learning reaction patterns like traditional methods, we tend to understand the content of conversation. Experimental results confirm the superiority of our framework over baseline scheme when we consider semantic information of a speech.

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